

DISCRETE DIFFERENTIAL GEOMETRY: AN APPLIED INTRODUCTION

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LECTURE 1: OVERVIEW

Discrete differential geometry (DDG) connects two ways of studying shape. Smooth geometry describes quantities such as tangent directions and bending. Geometry processing computes with finite data, such as polygon vertices and triangle meshes. DDG asks which smooth ideas can be kept in these finite models.

Applications include smoothing surfaces, comparing shapes, simulation, and analyzing biological or manufactured objects.

Definition: DIFFERENTIAL GEOMETRY is a language for describing local properties of shape, such as how fast a curve is traversed or how much a surface bends at a point. It also connects local quantities, such as bending at a point, to properties of the whole shape, such as the total turning of a closed curve.

Definition: DISCRETE DIFFERENTIAL GEOMETRY expresses these ideas using finite data: vertices, edges, lengths, angles, and areas. It replaces derivatives at every point with quantities that a computer can store and calculate.

1.1 The DDG translation problem

A smooth quantity can have several equivalent definitions. Apply each definition to a polygon or mesh, and they may give different answers. One may preserve an integral identity, another may be easier to optimize, and another may be more stable to compute. Compare them against the needs of the problem; there is usually no single best choice.

This is a design principle rather than a defect. A discrete definition should be judged against the job it must do: for example, convergence under refinement, preservation of a desired invariant, or computational cost.

Note. The most natural discrete analogue of a pointwise smooth quantity is often an *integrated* quantity. Integration replaces an infinitesimal rate with a finite change over an edge, face, or other cell.

1.2 Plane curves and their local frame

Let a regular plane curve be a smooth map

$$\gamma : [0, L] \longrightarrow \mathbb{R}^2.$$

When its parameter s is arc length, $\|\gamma'(s)\| = 1$ and the unit tangent is

$$T(s) = \gamma'(s).$$

Choose the unit normal $N(s)$ by rotating $T(s)$ counterclockwise by a quarter turn. The SIGNED CURVATURE $\kappa(s)$ is defined through

$$T'(s) = \kappa(s)N(s).$$

Thus curvature is a second-order quantity: it records how the direction of a curve changes per unit length. Its sign depends on the chosen orientation. If $\theta(s)$ is the angle of the tangent, then the same information is expressed by $\kappa = d\theta/ds$. Figure 1(a) shows the frame.

Example: For a circle of radius r with the normal oriented toward its center, $\kappa = 1/r$. Smaller circles bend more sharply, which matches the reciprocal dependence on radius.

1.3 Why a polygon needs a new definition

Consider an oriented polygon with vertices $x_0, \dots, x_m$. Write

$$e_i = x_i - x_{i-1}, \quad \ell_i = \|e_i\|, \quad T_i = \frac{e_i}{\ell_i}.$$

Along an edge the tangent is constant, so a literal second derivative is zero. At a vertex the tangent jumps, so the same derivative is singular. A pointwise copy of smooth curvature therefore cannot describe the polygon.

The first useful replacement is the signed TURNING ANGLE Θ_i from T_i to T_{i+1} (figure 1(b)). It is a finite version of the integrated smooth identity

$$\int_a^b \kappa(s) ds = \theta(b) - \theta(a).$$

For a polygon, the curvature concentrated at vertex x_i is naturally the turning angle Θ_i . This quantity has no units: it measures total turning, not bending per unit length.

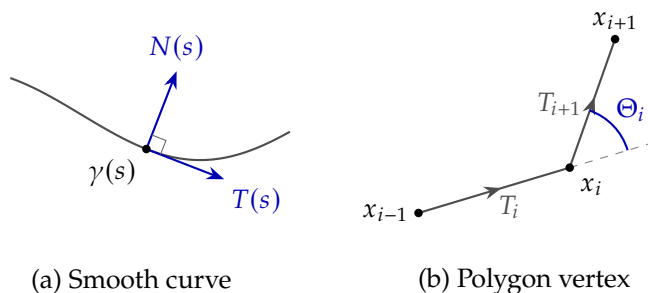


Figure 1: (a) The unit tangent $T(s)$ and the unit normal $N(s)$, obtained by a counterclockwise quarter turn of $T(s)$; here the curve bends toward N , so $\kappa > 0$. (b) On a polygon the tangent is constant along each edge and jumps at a vertex; the turning angle Θ_i measures that jump.

Definition 1.1 [INTEGRATED CURVATURE OF A POLYGON]. At an interior vertex x_i , the integrated curvature is the signed turning angle

$$\Theta_i = \text{atan2}(\det(T_i, T_{i+1}), \langle T_i, T_{i+1} \rangle).$$

For a consistently oriented simple closed polygon, the sum of these angles is 2π (with sign determined by orientation).

1.4 Curvature from variation of length

Curvature also controls how length changes under a normal deformation. For a smooth curve with a displacement field V , the first variation of length (ignoring endpoint terms) is

$$\delta L[V] = - \int \kappa \langle V, N \rangle ds.$$

Moving in the direction κN therefore decreases length most rapidly when the curve is free to move normally.

For a polygon, total length is $L(x) = \sum_i \ell_i$. Differentiating this finite expression with respect to an interior vertex gives

$$\nabla_{x_i} L = T_i - T_{i+1}.$$

The negative gradient $T_{i+1} - T_i$ is a DISCRETE CURVATURE-NORMAL VECTOR. Its magnitude is $2 \sin(|\Theta_i|/2)$, which agrees with the turning angle to first order for small angles. It points along the bisector of the angle at x_i , toward the side to which the polygon turns (figure 2(a)). Unlike Θ_i , this vector comes directly from minimizing length. It is therefore useful for smoothing a curve by gradient descent.

1.5 Other finite viewpoints

Two further smooth descriptions of curvature lead to more discrete definitions.

First, a normal offset of a smooth curve changes its length in proportion to integrated curvature. This is STEINER'S FORMULA:

$$\text{length}(\gamma + \varepsilon N) = \text{length}(\gamma) - \varepsilon \int \kappa ds.$$

A polygon has no single normal at a vertex, so each edge is offset along its own normal, and the gaps at the vertices must be closed by a choice: (A) a circular arc of radius ε , (B) a straight connector, or (C) extending the offset edges until they intersect (figure 2(b)). At a vertex with turning angle Θ_i these add $\varepsilon \Theta_i$, $2\varepsilon \sin(\Theta_i/2)$, and $2\varepsilon \tan(\Theta_i/2)$ of length, respectively, to the outer offset of a convex polygon. Matching the coefficient of ε with Steiner's formula gives three different finite length-change formulas, so the vertices alone do not determine one preferred definition.

Second, a smooth curve is locally approximated by an OSCULATING CIRCLE, whose curvature is the reciprocal of its radius. Three consecutive polygon vertices determine a circumcircle when they are not collinear (figure 2(c)), suggesting a curvature estimate $1/R$. With $w_i = \|x_{i+1} - x_{i-1}\|$, the law of sines gives

$$\frac{1}{R} = \frac{2 \sin \Theta_i}{w_i}.$$

This estimate emphasizes local circle fitting rather than an integral law or a variational property.

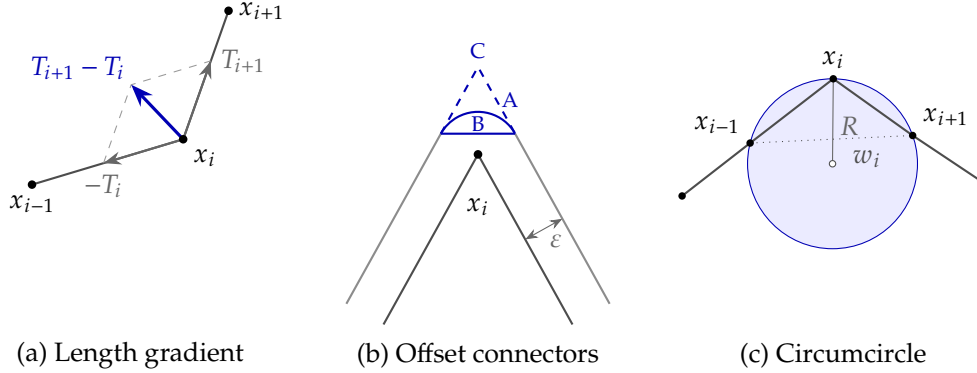


Figure 2: Three further discrete viewpoints at a vertex x_i . (a) The negative length gradient $T_{i+1} - T_i$ points along the angle bisector, with magnitude $2 \sin(|\Theta_i|/2)$. (b) Offsetting each edge by ε leaves a gap at the vertex, closed by an arc (A), a straight segment (B), or by extending the edges (C). (c) The circle through x_{i-1}, x_i, x_{i+1} has radius R with $1/R = 2 \sin \Theta_i / w_i$.

Note. Turning angle, length variation, normal offsets, and circumcircles agree for a sufficiently refined smooth curve, but they are generally inequivalent on a coarse polygon. Selecting among them is part of modeling the problem.

The four resulting curvatures at a vertex are summarized below. The first three are dimensionless and agree with Θ_i to first order as $\Theta_i \rightarrow 0$. The last has units of inverse length, like smooth curvature, and satisfies $(w_i/2) \kappa_i^D = \sin \Theta_i \approx \Theta_i$.

Name	Origin	Discrete curvature at x_i
κ_i^A	Turning angle; circular-arc offset	Θ_i
κ_i^B	Length gradient; straight-connector offset	$2 \sin(\Theta_i/2)$
κ_i^C	Extended-edge offset	$2 \tan(\Theta_i/2)$
κ_i^D	Circumcircle	$2 \sin(\Theta_i)/w_i$

1.6 A test case: curve shortening

CURVE SHORTENING FLOW evolves a curve by

$$\frac{\partial \gamma}{\partial t} = \kappa N.$$

More curved regions move faster. Smooth flow of a closed curve has three notable properties: total curvature stays constant, the center of mass does not drift, and, up to rescaling, circles remain circles. A discrete version moves each vertex a small step along its discrete curvature normal,

$$x_i^{t+1} = x_i^t + \tau \kappa_i N_i,$$

with step size τ . No choice of discrete curvature preserves all three properties at once; in fact, it is impossible. Using κ^A keeps total curvature fixed, κ^B (the length gradient) avoids drift, and κ^D keeps a circular polygon round, but each fails the other two tests. Choose the properties needed for the application.

Source: Keenan Crane, Discrete Differential Geometry: An Applied Introduction, CMU 15-458/858, Lecture 1: "Overview," official slides, <https://brickisland.net/ddg-web/lectures/DDG-Overview.pdf> (accessed August 4, 2026).